

MARQUETTE COUNTY AP, MI, USA

WMO: 727430

Lat: 46.5310N

Lon: 87.5490W

Elev: 431

StdP: 96.25

Time Zone: -5.00 (NAE)

Period: 90-99

WBAN: 94850

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-22.7	-31.1	0.2	-25.5	-27.4	0.3	-21.8	11.1	-9.5	9.9	-10.4	2.8	270	0.686

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	28.8	20.0	27.0	18.7	25.2	17.8	21.6	26.9	20.3	24.7	19.3	23.4	5.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	15.4	23.9	18.7	14.3	23.1	17.7	13.3	22.0	64.7	27.3	60.4	24.7	56.7	23.5	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.4	7.9	-30.8	32.7	3.8	1.8	-33.5	34.0	-35.7	35.1	-37.8	36.1	-40.5	37.4		
(5)			-30.9	23.8	3.7	1.6	-33.6	24.9	-35.8	25.8	-37.9	26.7	-40.6	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-10.3	-8.7	-4.5	2.2	9.5	15.3	17.0	16.5	12.1	6.1	-1.8	-6.4		
	DBStd	11.12	6.27	6.90	6.37	5.84	5.49	4.95	3.67	3.91	4.83	4.64	4.76	6.08		
	HDD10.0	2978	631	523	451	246	79	11	1	1	32	141	356	508		
	HDD18.3	5337	889	756	708	486	281	114	72	82	196	381	605	766		
	CDD10.0	778	0	0	1	12	63	171	217	203	93	18	1	0		
	CDD18.3	95	0	0	0	2	6	24	30	27	7	0	0	0		
	CDH23.3	842	0	0	0	16	77	226	259	204	54	6	0	0		
	CDH26.7	184	0	0	0	5	13	55	60	39	11	1	0	0		
(14) Wind	WSAvg	4.0	4.3	4.3	4.1	4.1	3.9	3.5	3.6	3.5	4.0	4.4	4.4	4.3		
(15) Precipitation	PrecAvg	915	62	44	74	67	79	87	76	85	99	91	79	64		
	PrecMax	1310	168	94	154	167	201	311	142	218	193	193	210	176		
	PrecMin	651	16	14	8	23	2	16	14	14	31	24	25	13		
	PrecStd	161	36	22	43	34	42	56	38	41	45	40	43	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	8.3	14.1	25.1	28.4	31.5	31.2	30.1	27.9	23.1	14.0	8.0		
		MCWB	1.2	3.3	8.9	15.7	17.0	23.2	21.4	20.4	19.4	15.2	12.4	4.0		
	2%	DB	1.0	4.9	10.1	19.2	25.5	28.0	28.0	27.5	24.7	19.0	9.4	4.3		
		MCWB	-0.3	1.9	5.9	11.3	15.8	19.7	20.0	19.6	18.2	13.6	6.9	1.5		
	5%	DB	-0.5	2.6	7.4	15.1	22.7	26.0	26.3	25.9	22.2	16.3	6.9	2.3		
		MCWB	-1.3	0.2	4.1	9.3	14.4	18.0	18.7	18.7	17.0	12.5	4.7	0.6		
	10%	DB	-2.0	0.9	4.3	11.8	19.5	23.9	24.1	23.8	19.8	13.8	4.8	1.0		
		MCWB	-2.8	-0.6	1.6	6.8	13.1	17.0	18.1	17.9	15.9	10.4	2.8	-0.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	4.4	10.2	17.5	19.7	23.5	23.3	22.4	20.8	16.6	12.4	4.7		
		MCDB	3.2	7.4	12.5	24.5	24.7	30.0	28.5	27.6	26.2	20.6	14.6	7.5		
	2%	WB	-0.1	1.8	7.3	12.7	17.2	21.3	21.6	20.8	19.0	14.8	7.6	2.1		
		MCDB	0.8	4.1	9.5	16.3	22.8	26.6	26.5	24.8	23.0	17.9	8.7	3.9		
	5%	WB	-1.2	0.4	4.1	9.5	15.5	19.6	20.1	20.0	17.7	12.9	4.9	0.7		
		MCDB	-0.4	2.3	7.3	15.1	21.4	24.0	24.6	23.7	21.3	16.3	6.6	2.0		
	10%	WB	-2.6	-0.8	1.6	6.8	13.5	18.0	18.9	18.9	16.5	10.3	3.0	-0.4		
		MCDB	-1.9	0.7	4.3	11.6	18.6	22.1	23.0	22.5	19.2	13.7	4.4	0.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	10.0	10.8	10.1	12.2	12.0	11.4	11.5	10.7	9.3	6.9	7.2		
		MCDBR	8.8	11.3	13.0	16.4	15.7	14.2	14.1	13.1	13.4	14.0	9.6	8.3		
	5% WB	MCWBR	8.0	8.9	9.2	9.5	8.4	7.8	7.6	7.0	8.1	9.5	7.7	6.3		
		MCWBR	8.5	9.6	11.7	14.5	13.5	11.8	12.2	10.8	11.3	12.5	9.1	7.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.256	0.270	0.302	0.329	0.363	0.364	0.370	0.357	0.346	0.318	0.290	0.256		
		taud	2.433	2.381	2.387	2.416	2.392	2.412	2.423	2.457	2.490	2.546	2.539	2.486		
	Ebn at Noon	Ebn at Noon	862	914	928	929	903	900	890	889	870	850	814	823		
		Edh at Noon	67	86	101	108	115	113	111	103	91	74	61	57		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.05	1.89	3.33	4.45	5.44	5.89	6.01	5.09	3.76	2.22	1.18	0.81		
	RadStd	RadStd	0.08	0.25	0.31	0.47	0.39	0.45	0.31	0.32	0.25	0.26	0.15	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page